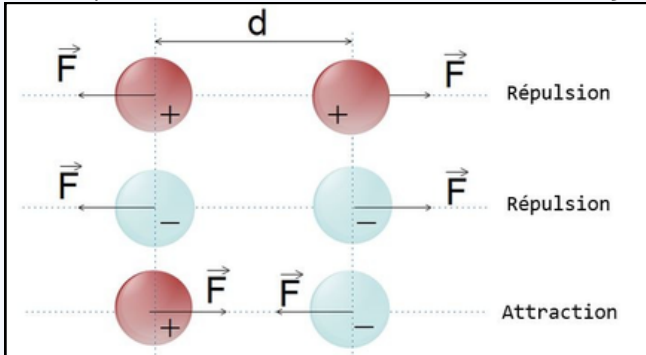


TECHNICAL SHEET (FOR EXPERTS / TEACHERS) – WORKSHOP 4 : ELECTROSTATICS & THE TRIBOELECTRIC EFFECT

The Repulsion and Attraction of Electric Charges



Static Electricity with a Balloon



Background : This educational workshop demonstrates the accumulation of electric charges at rest on the surface of insulating materials and the interaction between them at a distance via an electric field.

Objectives of the experiment : To demonstrate the removal of electrons through mechanical friction and to use the Coulomb force to counteract gravity by making pieces of tissue paper float away.

I. Physical Phenomena in Focus

- Friction-induced electrification (triboelectric effect) : Mechanical friction between the wool band and the PVC tube transfers thermal energy to the outer electrons. According to the triboelectric series, wool releases its electrons (overall positive charge +), while PVC, which is highly electronegative, captures these charges (overall negative charge -).
- Coulomb's Law and the electrostatic field : The charged PVC tube accumulates a net surface charge of $-Q$. It generates an electric field $\mathbf{E}$ in the surrounding space. The electrostatic force experienced by a charged body q placed in this field is given by : $\mathbf{F} = q \cdot \mathbf{E}$
- Induced polarization (dielectrics) : When the tube approaches the tissue paper (an insulating/neutral material), the electric field $\mathbf{E}$ induces a microscopic displacement of charges within the paper's atoms (polarization). The positive charges in the paper are attracted toward the negatively charged PVC. If the attractive Coulomb force is greater than the weight of the sheet ($F > m \cdot g$), the paper flies away.

II. Experimental Protocol and Technical Alignment

1. Controlling ambient humidity : Water in the air makes the surface of materials conductive, instantly dissipating accumulated charges to the ground. The 20-cm PVC tube must be completely dry.
2. Directional friction : Rub vigorously in one direction with the wool cloth to maximize the surface charge density (σ).

DID YOU KNOW THAT ?

Did you know ? Harnessing the invisible power of lightning ! By rubbing our PVC tube with wool, you've just dislodged billions of tiny, invisible electric particles: electrons. The tube then turns into a real magnet for electricity (negatively charged). When you bring it close to the tissue paper, it creates an invisible force capable of overcoming Earth's gravity and making the paper fly away ! This is called **electrostatics**. It's the same physical phenomenon that makes your hair stand on end when you take off a sweater in winter, or that builds up enormous amounts of energy in clouds just before lightning strikes the ground ! (Academic source: IEEE Transactions on Dielectrics and Electrical Insulation).